ASS 2

1. Link: https://github.com/Kur00o/CNL/blob/main/ASS2.c

```c
// ASS 2 - hamming code


#include <stdio.h>
#include <math.h>


int main() {
   int data[20], hamming[30];
   int m, r = 0;


   printf("Enter number of data bits: ");
   scanf("%d", &m);


   printf("Enter data bits:\n");
   for (int i = 1; i <= m; i++)
       scanf("%d", &data[i]);


   while ((pow(2, r)) < (m + r + 1))
       r++;


   int total = m + r;


   int j = 1;
   for (int i = 1; i <= total; i++) {
       if ((i & (i - 1)) == 0)
           hamming[i] = 0;
       else
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        hamming[i] = data[j++];
}

for (int i = 0; i < r; i++) {
    int pos = pow(2, i);
    int parity = 0;

    for (int j = 1; j <= total; j++) {
        if (j & pos)
            parity ^= hamming[j];
    }

    hamming[pos] = parity;
}

printf("\nTransmitted Hamming Code:\n");
for (int i = total; i >= 1; i--)
    printf("%d ", hamming[i]);

printf("\n");

int received[30];

printf("\nEnter the received Hamming code (position 1 to %d):\n", total);
for (int i = 1; i <= total; i++)
{
    scanf("%d", &received[i]);
}

int error = 0;
```

```
for (int i = 0; i < r; i++)
{
    int pos = (int)pow(2, i);
    int parity = 0;

    for (int j = 1; j <= total; j++)
    {
        if (j & pos)
            parity ^= received[j];
    }

    if (parity)
        error += pos;
}

if (error == 0)
{
    printf("\nNo error detected.\n");
}
else
{
    printf("\nError detected at position %d.\n", error);

    received[error] ^= 1;

    printf("Corrected Hamming Code:\n");
    for (int i = total; i >= 1; i--)
        printf("%d ", hamming[i]);
```

```
        printf("\n");
    }


    return 0;
}
```

OUTPUT

